# KIRUTHIKA M

Electronics &CommunicationEngineering
Embedded Systems | IoT
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LinkedIn Profile

## CAREER OBJECTIVE

Enthusiastic ECE fresher seeking an entry-level role in Embedded Systems or IoT to leverage academic knowledge in hardware design, sensor integration, and firmware development while adding value to an innovative organisation.

## TECHNICAL SKILLS

**Core Skills:** Embedded Systems, IoT, Verilog HDL, Digital Electronics, VLSI Design, Functional Verification (UVM)

**Tools & Platforms:** Arduino, ESP32, XIAO ESP32-S3, ESP32-CAM, Sensors, Proteus Simulation, Circuit Design, EDA Playground, Blynk

**Languages:** Verilog HDL  Other: HTML, CSS (Basics)

## EDUCATION

**B.E. – Electronics and Communication Engineering** 2023 – 2027
KSR Institute for Engineering and Technology, Tiruchengode | CGPA: 7.91 (Till 5th Semester)

**Higher Secondary School (12th Grade)** 2023
Government Girls Higher Secondary School, Tiruchengode | Percentage: 80%

**Secondary School (10th Grade)**
Naladas Matric and Higher Secondary School, Tiruchengode 2020

## PROJECTS

**Smart Hand Glove – XIAO ESP32-S3**

- Developed a wearable system for real-time gesture recognition; implemented sensor-based data processing for intuitive human-device interaction.

**Image Processing-Based Fire Detection – ESP32-CAM**

- Built a fire detection system using real-time image processing and integrated automated alerts for remote safety monitoring.

**VLSI Design & Functional Verification**

- Designed RTL modules in Verilog HDL; performed functional verification using UVM methodologies with testbench development.

## INTERNSHIPS & TRAINING

**IoT & Cloud Computing Internship – Samcore Technology, Tiruchirappalli**

- Covered IoT concepts, cloud integration, device connectivity, and real-time data transmission principles.

**Web Development Internship – Mayra Software Solutions, Erode**

- Gained fundamentals of front-end development; built simple web applications using HTML and CSS.

## ACHIEVEMENTS & CERTIFICATIONS

- Volunteered at an IoT workshop, assisting participants with sensor integration and Blynk visualisation.
- Completed VLSI Design & Functional Verification certification – RTL design and UVM testbench development.